

Lab 1: Requirements Engineering & UML

Use-Case Modelling

Problem Statement #31: Multi-Vendor Artisan E-Commerce Marketplace

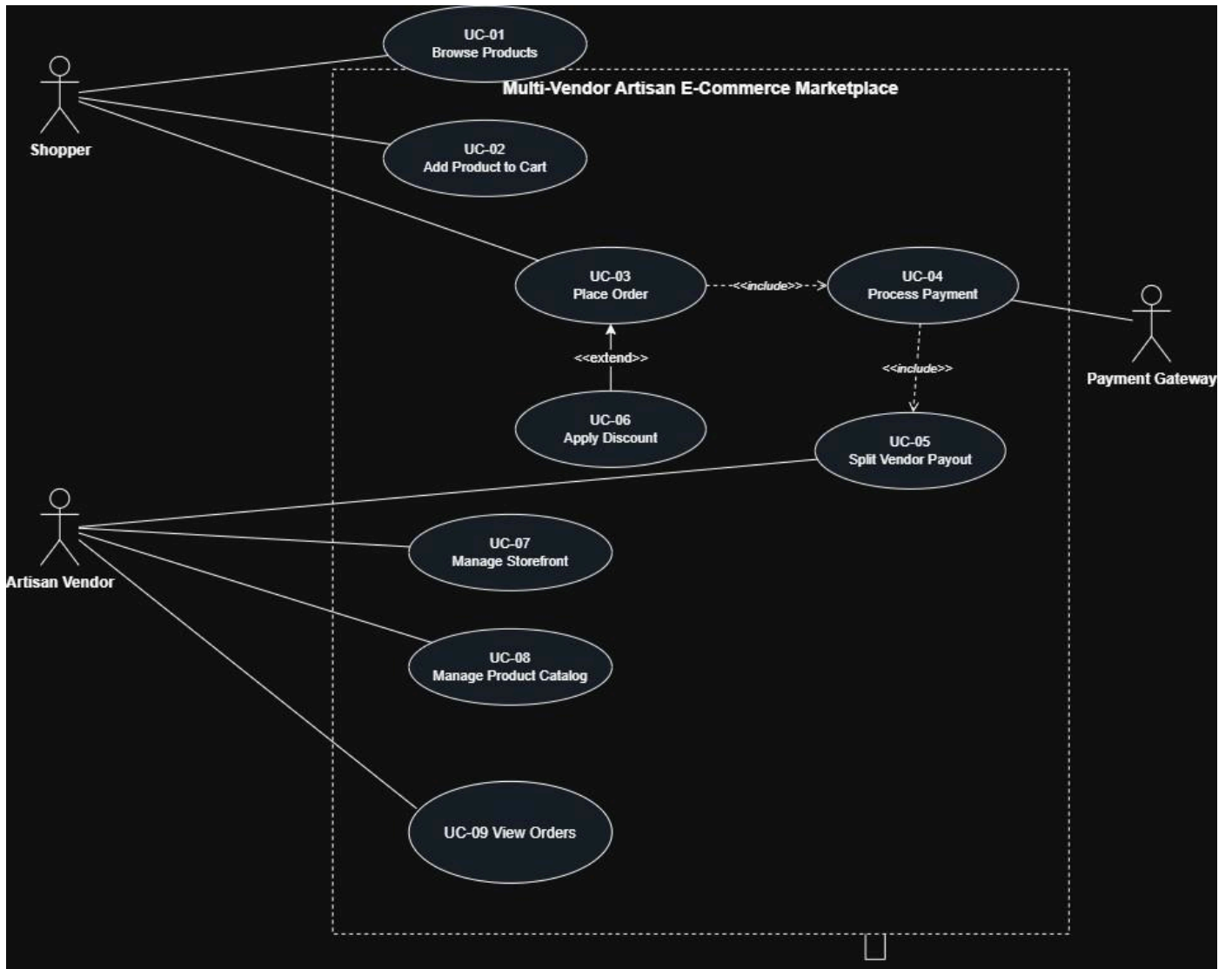
Req ID	Type	Description	Priority	Acceptance Criteria	Rationale
FR-001	Functional	The system shall automatically divide a customer's cart payment among the respective artisan vendors after deducting a 5% platform commission	High	Pass: The sum of all vendor payouts and the 5% platform commission equals the customer's total cart payment. Fail: The calculated amounts do not equal the customer's total cart payment.	Ensures correct payment distribution to vendors and collection of the platform commission.
FR-002	Functional	The system shall allow an artisan vendor to create and manage an online storefront	High	Pass: An artisan vendor can create a storefront and successfully update its information. Fail: The	Allows artisans to establish and maintain their presence on the marketplace.

				storefront cannot be created or updated.	
FR-003	Functional	The system shall allow artisan vendors to add, update, and remove products from their product catalog.	High	Pass: An artisan vendor can add, update, and remove a product, and each change is reflected in the product catalog. Fail: Any of the add, update, or remove operations cannot be completed successfully.	Allows vendors to maintain accurate and up-to-date product information.
FR-004	Functional	The system shall allow shoppers to browse products from different artisan vendors and add selected products to a shopping cart.	High	Pass: A shopper can browse products from multiple vendors and add selected products to the shopping cart. Fail: The shopper cannot browse products or add products to the shopping cart.	Provides the core shopping functionality of the marketplace.

	Functional	The system shall allow shoppers to place orders and receive an	High	Pass: After successful payment, an order is created and an order	Completes the purchasing process for the shopper.
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		order confirmation after successful payment.		confirmation is displayed to the shopper. Fail: The payment succeeds but the order or confirmation is not generated.	
NFR-001	Non Func	The system shall support high-resolution product image rendering using CDN caching with a catalog loading time below 500 ms.	High	Pass: Performance testing confirms that the product catalog loads in less than 500 ms under simulated peak load. Fail: The measured catalog loading time is 500 ms or higher.	Ensures fast and responsive product browsing for shoppers.

NFR-002	Non Func	The system shall maintain continuous availability of the marketplace for shoppers and artisan vendors.	High	Pass: Availability monitoring confirms that the marketplace is accessible for at least 99.9% of the scheduled operating period. Fail: Availability falls below 99.9%.	Ensures shoppers and vendors can access the marketplace reliably.
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Use-Case Flow Specification

Problem Statement #31 — Multi-Vendor Artisan E-Commerce Marketplace

Use Case: UC-03 Place Order

Actors: Shopper (primary), Payment Gateway (secondary/supporting)

Preconditions:

1. Shopper has an active account and is logged in.
2. Shopping cart contains at least one product from one or more artisan vendors.
3. Selected products are in stock and available for purchase.

Postconditions:

1. An order record is created and confirmed in the system.
2. Payment has been captured and split across the relevant artisan vendor accounts, less the 5% platform commission.
3. Shopper receives an order confirmation; the shopping cart is cleared.

Main Success Scenario:

1. Shopper reviews the shopping cart and selects "Place Order".
2. System displays the order summary, including itemized costs per vendor and total amount due.
3. Shopper confirms the shipping address and selects a payment method.
4. System invokes UC-04 Process Payment «include» to charge the shopper's chosen payment method via the Payment Gateway.
5. Payment Gateway authorizes the transaction and returns a success response.
6. System invokes UC-05 Split Vendor Payout «include» to allocate earnings to each artisan vendor's account after deducting a 5% platform commission.
7. System creates the order record and updates inventory for each purchased product.
8. System displays an order confirmation with order ID and estimated delivery details, and emails a receipt to the shopper.
9. Use case ends successfully.

Alternate Flow:

5a. Payment Declined

- 5a1. Payment Gateway returns a declined/failed status to the system.
- 5a2. System displays a "Payment Failed" message and prompts the shopper to retry or choose a different payment method.
- 5a3. If the shopper selects a new payment method, the flow resumes at step 4.
- 5a4. If payment fails after two attempts, the system cancels the order attempt, notifies the shopper, and the shopping cart is preserved for a later retry. Use case ends in failure.